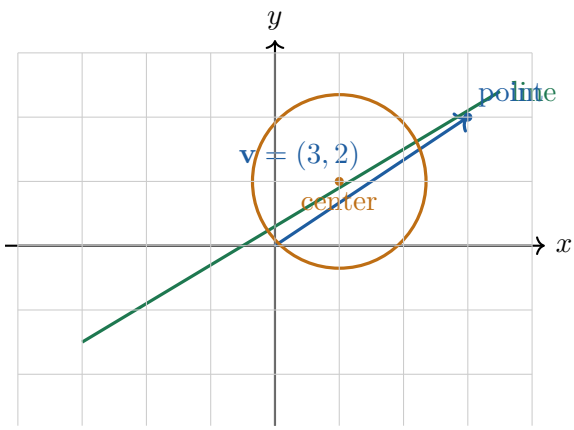
Analytic Geometry for Absolute Beginners

Lecture Notes for Computer Scientists

Prepared as a slow, step-by-step lecture script

May 16, 2026



Key idea

Analytic geometry is the art of turning pictures into numbers and equations. A point becomes a pair of numbers. A line becomes an equation. A geometric question becomes an algebraic calculation. This is exactly why analytic geometry is useful in programming, graphics, games, robotics, machine learning, computer vision, and computational geometry.

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How to read these notes

These notes are intentionally slow. They do not assume that the reader already feels comfortable with geometry. The goal is not to impress anyone with advanced notation. The goal is to make the basic ideas usable.

Every important idea is introduced in four stages:

1. What the object means in a picture.
2. How we write it with numbers.
3. How to compute with it.
4. What mistakes beginners usually make.

For a computer science student, the key message is simple:

> If geometry can be written as coordinates, vectors, matrices, and equations, then geometry can be implemented in code.

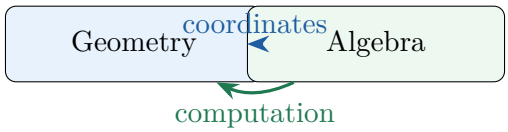
1 Why analytic geometry exists

1.1 Geometry before coordinates

Classical geometry studies shapes: points, lines, circles, triangles, distances, angles, and intersections. In a purely visual approach we draw a picture and reason about it.

That is useful, but it has a weakness: a computer cannot directly reason from a hand-drawn picture. A computer needs numbers and rules.

Analytic geometry solves this problem. It connects geometry with algebra.



Key idea

Analytic geometry is a dictionary:

Geometry	becomes	Algebra
point	becomes	coordinate pair (x, y)
line	becomes	equation $ax + by + c = 0$
circle	becomes	equation $(x - a)^2 + (y - b)^2 = r^2$
intersection	becomes	solving equations
distance	becomes	a formula with square roots
movement	becomes	vector addition
rotation/scaling	becomes	matrix multiplication

1.2 Why computer scientists need it

Analytic geometry appears in many areas of computing:

- computer graphics: moving, rotating, scaling objects;
- games: collision detection, camera movement, rays and intersections;
- robotics: position, orientation, distance to obstacles;
- machine learning: vectors, distances, projections, hyperplanes;
- computer vision: pixels, coordinates, lines, camera geometry;
- algorithms: points, polygons, convex hulls, nearest neighbors.

A simple example: suppose a game character is at position $(2, 1)$ and moves by vector $(3, 4)$. The new position is

$$(2, 1) + (3, 4) = (5, 5).$$

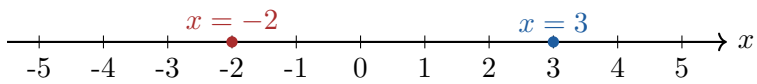
That is analytic geometry in its simplest computational form.

2 The coordinate plane from zero

2.1 The number line

Before we use two coordinates, we need one coordinate.

A number line is a straight line with numbers on it. Each point on the line corresponds to a real number.



The distance between -2 and 3 is

$$|3 - (-2)| = |5| = 5.$$

Common mistake

The distance between two numbers is never negative. That is why we use absolute value:

$$\text{distance on a line} = |x_2 - x_1|.$$

2.2 Two coordinates

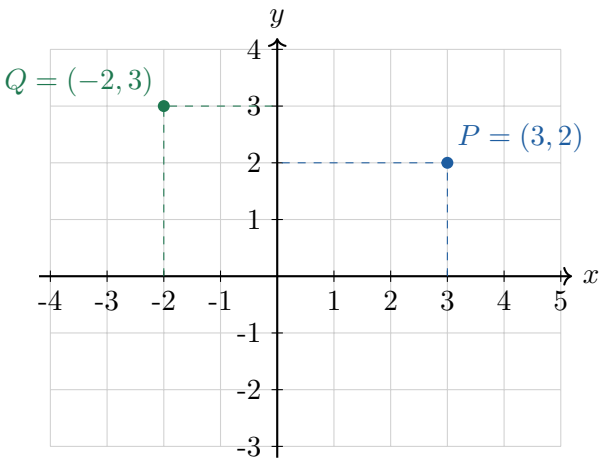
The coordinate plane uses two number lines:

- the horizontal axis, called the x-axis;
- the vertical axis, called the y-axis.

A point is written as

$$P = (x, y).$$

The first number tells us how far to move horizontally. The second number tells us how far to move vertically.



Recipe

To plot the point $(3, 2)$:

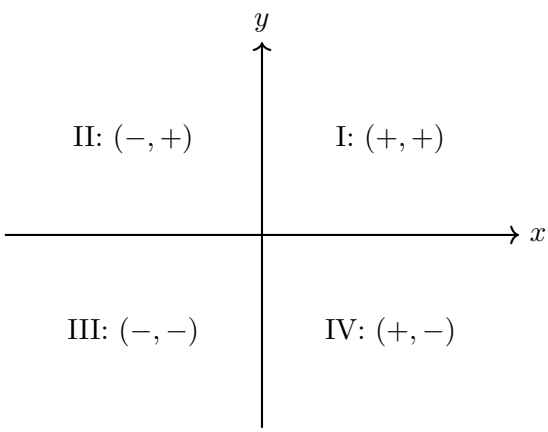
1. Start at the origin $(0, 0)$.
2. Move 3 units to the right.
3. Move 2 units up.
4. Mark the point.

Common mistake

Do not swap coordinates. The point $(3, 2)$ is not the same as $(2, 3)$. The first number is always horizontal. The second number is always vertical.

2.3 Quadrants

The coordinate axes divide the plane into four regions called quadrants.



Quick practice

Say in which quadrant each point lies:

1. $(4, 1)$
2. $(-3, 2)$
3. $(-5, -1)$
4. $(2, -7)$

3 Distance and midpoint

3.1 Distance between two points

Let

$$A = (x_1, y_1), \qquad B = (x_2, y_2).$$

The horizontal change is

$$\Delta x = x_2 - x_1.$$

The vertical change is

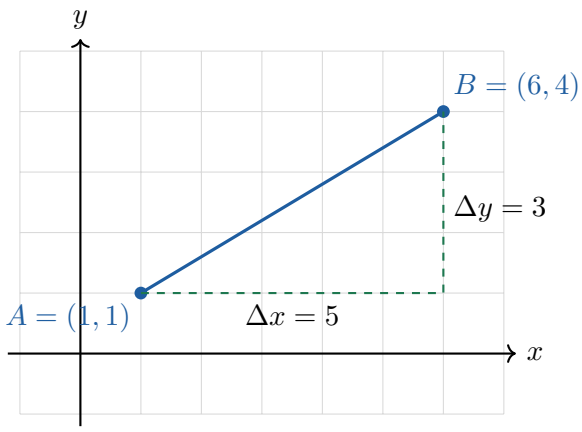
$$\Delta y = y_2 - y_1.$$

These two changes form a right triangle. By the Pythagorean theorem,

$$AB^2 = (\Delta x)^2 + (\Delta y)^2.$$

Therefore the distance is

$$\boxed{d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.}$$



Example

Find the distance between $A = (1, 1)$ and $B = (6, 4)$.
Step 1: compute coordinate differences:

$$\Delta x = 6 - 1 = 5, \qquad \Delta y = 4 - 1 = 3.$$

Step 2: use the formula:

$$d = \sqrt{5^2 + 3^2} = \sqrt{25 + 9} = \sqrt{34}.$$

So the exact distance is $\sqrt{34} \approx 5.83$.

Common mistake

Never write

$$\sqrt{x_2 - x_1 + y_2 - y_1}.$$

That is not the distance formula. The differences must be squared before adding.

3.2 Midpoint of a segment

The midpoint is exactly halfway between two points.

If

$$A = (x_1, y_1), \qquad B = (x_2, y_2),$$

then the midpoint is

$$\boxed{M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right).}$$

Example

Find the midpoint of $A = (2, 5)$ and $B = (8, 1)$.

$$M = \left(\frac{2+8}{2}, \frac{5+1}{2}\right) = (5, 3).$$

Key idea

The midpoint formula is just averaging. Average the x-coordinates. Average the y-coordinates.

Quick practice

Compute the distance and midpoint for each pair:

1. $A = (0, 0)$, $B = (3, 4)$
2. $A = (-1, 2)$, $B = (5, 2)$
3. $A = (-3, -2)$, $B = (1, 4)$

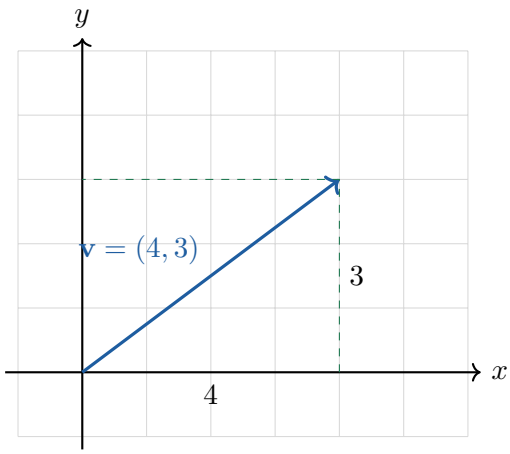
4 Vectors: arrows that can be computed

4.1 What is a vector?

A vector is an object with direction and length. In the coordinate plane we usually write a vector as an ordered pair:

$$\mathbf{v} = (a, b).$$

This means: move a units horizontally and b units vertically.



4.2 Vector from one point to another

If $A = (x_1, y_1)$ and $B = (x_2, y_2)$, then the vector from A to B is

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1).$$

Example

Let $A = (2, 1)$ and $B = (7, 4)$. Then

$$\overrightarrow{AB} = (7 - 2, 4 - 1) = (5, 3).$$

This means: to move from A to B, go 5 units right and 3 units up.

Common mistake

The vector from A to B is not the same as the vector from B to A:

$$\overrightarrow{BA} = -(\overrightarrow{AB}).$$

Direction matters.

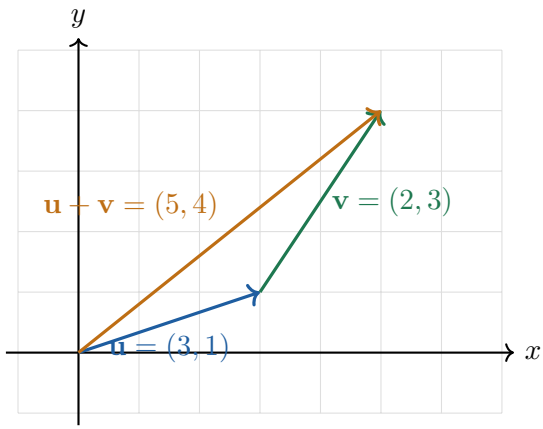
4.3 Adding vectors

If

$$\mathbf{u} = (a, b), \qquad \mathbf{v} = (c, d),$$

then

$$\mathbf{u} + \mathbf{v} = (a + c, b + d).$$



Example

Let $\mathbf{u} = (3, -2)$ and $\mathbf{v} = (-1, 5)$. Then

$$\mathbf{u} + \mathbf{v} = (3 + (-1), -2 + 5) = (2, 3).$$

4.4 Multiplying a vector by a number

If $\mathbf{v} = (a, b)$ and k is a number, then

$$k\mathbf{v} = (ka, kb).$$

Multiplication by k changes the length of the vector. If $k < 0$, it also reverses the direction.

Example

If $\mathbf{v} = (2, 1)$, then

$$3\mathbf{v} = (6, 3), \qquad -2\mathbf{v} = (-4, -2).$$

4.5 Length of a vector

The length of $\mathbf{v} = (a, b)$ is

$$\boxed{\|\mathbf{v}\| = \sqrt{a^2 + b^2}.}$$

This is just the distance from $(0, 0)$ to (a, b).

Example

For $\mathbf{v} = (3, 4)$,

$$\|\mathbf{v}\| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5.$$

4.6 Unit vectors

A unit vector has length 1. To convert a nonzero vector into a unit vector in the same direction, divide by its length:

$$\widehat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Example

Let $\mathbf{v} = (3, 4)$. Its length is 5, so a unit vector in the same direction is

$$\widehat{\mathbf{v}} = \left(\frac{3}{5}, \frac{4}{5}\right).$$

Check:

$$\sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = 1.$$

Quick practice

For each vector, compute its length. If possible, find a unit vector in the same direction.

1. $(1, 0)$
2. $(0, -5)$
3. $(6, 8)$
4. $(-2, 2)$

5 The dot product and angles

5.1 Definition of the dot product

For two vectors

$$\mathbf{u} = (a, b), \qquad \mathbf{v} = (c, d),$$

the dot product is

$$\boxed{\mathbf{u} \cdot \mathbf{v} = ac + bd.}$$

This looks simple, but it is one of the most important formulas in analytic geometry.

Example

If $\mathbf{u} = (2, 3)$ and $\mathbf{v} = (5, -1)$, then

$$\mathbf{u} \cdot \mathbf{v} = 2 \cdot 5 + 3 \cdot (-1) = 10 - 3 = 7.$$

5.2 Geometric meaning

The dot product is connected with the angle between vectors:

$$\boxed{\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta.}$$

Therefore,

$$\boxed{\cos\theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}.}$$

Key idea

The dot product measures how much two vectors point in the same direction.

- Positive dot product: angle is acute, directions are similar.
- Zero dot product: angle is 90°, vectors are perpendicular.
- Negative dot product: angle is obtuse, directions are opposite-ish.

5.3 Perpendicular vectors

Two nonzero vectors are perpendicular if their dot product is zero:

$$\mathbf{u} \perp \mathbf{v} \quad \Longleftrightarrow \quad \mathbf{u} \cdot \mathbf{v} = 0.$$

Example

Are $\mathbf{u} = (3, 2)$ and $\mathbf{v} = (4, -6)$ perpendicular?

$$\mathbf{u} \cdot \mathbf{v} = 3 \cdot 4 + 2 \cdot (-6) = 12 - 12 = 0.$$

Yes, they are perpendicular.

Common mistake

A dot product is a number, not a vector. For example,

$$(1, 2) \cdot (3, 4) = 11.$$

The answer is not $(3, 8)$.

6 Lines in the plane

6.1 Slope: the steepness of a line

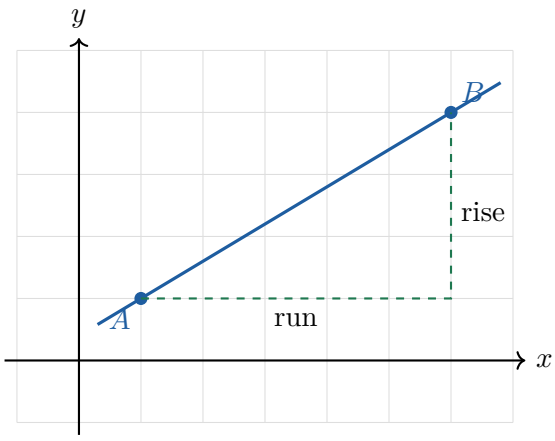
The slope of a nonvertical line measures how much y changes when x increases by 1.

For two points

$$A = (x_1, y_1), \qquad B = (x_2, y_2),$$

the slope is

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}.}$$



Example

Find the slope of the line through $A = (1, 1)$ and $B = (6, 4)$.

$$m = \frac{4-1}{6-1} = \frac{3}{5}.$$

This means: for every 5 units right, the line goes 3 units up.

Common mistake

The formula is

$$m = \frac{\text{change in } y}{\text{change in } x}.$$

Students often invert it by accident. Slope is rise over run, not run over rise.

6.2 Equation of a line: slope-intercept form

A common equation of a line is

$$\boxed{y = mx + b.}$$

Here:

- m is the slope;
- b is the y-intercept, the value of y when $x = 0$.

Example

The line

$$y = 2x + 3$$

has slope 2 and y-intercept 3. This means it crosses the y-axis at $(0, 3)$.

6.3 Point-slope form

If a line has slope m and passes through (x_0, y_0), then

$$\boxed{y - y_0 = m(x - x_0).}$$

Example

Find the line with slope 3 passing through $(2, 5)$.

$$y - 5 = 3(x - 2).$$

Now simplify:

$$y - 5 = 3x - 6,$$
$$y = 3x - 1.$$

6.4 Line through two points

To find a line through two points:

1. Compute the slope.
2. Use point-slope form.
3. Simplify if needed.

Example

Find the line through $A = (1, 2)$ and $B = (4, 8)$.
First compute the slope:

$$m = \frac{8 - 2}{4 - 1} = \frac{6}{3} = 2.$$

Use point-slope form with $A = (1, 2)$:

$$y - 2 = 2(x - 1).$$

Simplify:

$$y - 2 = 2x - 2,$$
$$y = 2x.$$

6.5 Horizontal and vertical lines

A horizontal line has equation

$$y = c.$$

Example:

$$y = 3.$$

Every point on this line has y-coordinate equal to 3.

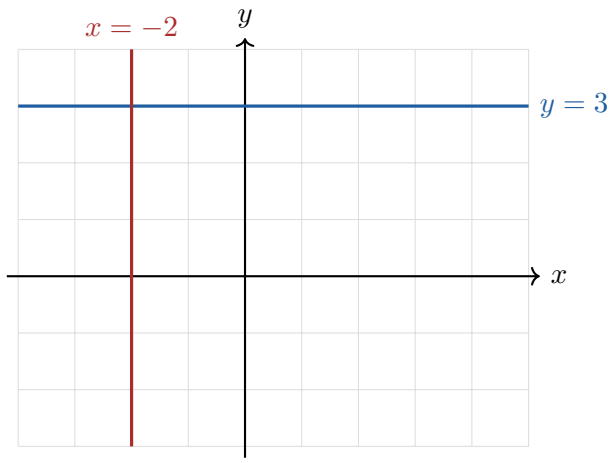
A vertical line has equation

$$x = c.$$

Example:

$$x = -2.$$

Every point on this line has x-coordinate equal to -2.



Common mistake

A vertical line does not have a slope. If you try to compute the slope, the denominator becomes zero:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{something}}{0}.$$

Division by zero is not allowed.

6.6 General equation of a line

Many formulas in analytic geometry use the general line equation

$$\boxed{ax + by + c = 0.}$$

This form is useful because it also includes vertical lines.

Example

The line

$$2x + 3y - 6 = 0$$

can be written in slope-intercept form:

$$3y = -2x + 6,$$

$$y = -\frac{2}{3}x + 2.$$

So its slope is $-\frac{2}{3}$ and its y-intercept is 2.

Quick practice

Find the equation of the line:

1. with slope 2 through $(1, 4)$;
2. through $(0, 1)$ and $(3, 7)$;
3. horizontal through $(5, -2)$;
4. vertical through $(5, -2)$.

7 Parallel and perpendicular lines

7.1 Parallel lines

Two nonvertical lines are parallel if they have the same slope.

If

$$y = m_1 x + b_1, \qquad y = m_2 x + b_2,$$

then the lines are parallel when

$$\boxed{m_1 = m_2.}$$

Example

The lines

$$y = 3x + 1, \qquad y = 3x - 5$$

are parallel because both have slope 3.

7.2 Perpendicular lines

Two nonvertical lines are perpendicular if their slopes multiply to -1:

$$\boxed{m_1 m_2 = -1.}$$

Equivalently,

$$\boxed{m_2 = -\frac{1}{m_1}.}$$

Example

A line has slope 2. What is the slope of a perpendicular line?

$$m_\perp = -\frac{1}{2}.$$

Example

Are the lines

$$y = \frac{3}{4}x + 1, \qquad y = -\frac{4}{3}x + 2$$

perpendicular?
Multiply slopes:

$$\frac{3}{4} \cdot \left(-\frac{4}{3}\right) = -1.$$

Yes, they are perpendicular.

Common mistake

Perpendicular slope means negative reciprocal, not just negative. The negative of 2 is -2, but the negative reciprocal of 2 is $-\frac{1}{2}$.

8 Intersections: geometry becomes algebra

8.1 What is an intersection?

An intersection point lies on both objects at the same time. For two lines, an intersection point must satisfy both line equations.

Therefore, finding the intersection of lines means solving a system of equations.

Example

Find the intersection of

$$y = x + 1, \qquad y = -2x + 7.$$

At the intersection, both expressions for y are equal:

$$x + 1 = -2x + 7.$$

Move terms:

$$3x = 6,$$

so

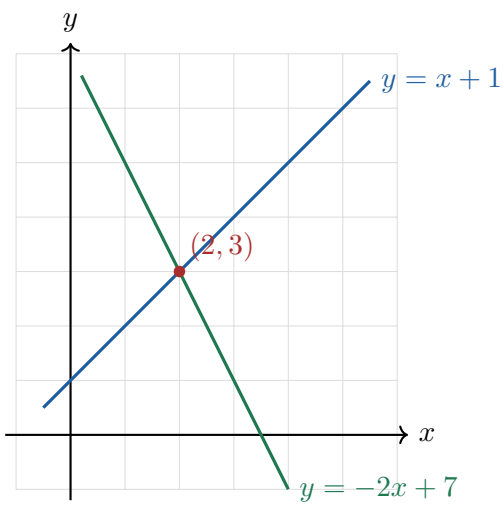
$$x = 2.$$

Substitute into $y = x + 1$:

$$y = 3.$$

The intersection point is

$$(2, 3).$$



8.2 Three possibilities

Two lines in the plane can have:

1. exactly one intersection point;
2. no intersection point, if they are parallel and distinct;
3. infinitely many intersection points, if they are the same line.

Key idea

In programming terms, line intersection is a branching problem. Your algorithm must handle all cases, not only the nice case where the lines cross once.

9 Distance from a point to a line

9.1 The formula

Let the line be

$$ax + by + c = 0,$$

and let the point be

$$P = (x_0, y_0).$$

The distance from P to the line is

$$\boxed{d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}.}$$

Example

Find the distance from $P = (1, 2)$ to the line

$$3x + 4y - 10 = 0.$$

Use the formula:

$$d = \frac{|3 \cdot 1 + 4 \cdot 2 - 10|}{\sqrt{3^2 + 4^2}} = \frac{|3 + 8 - 10|}{5} = \frac{1}{5}.$$

9.2 Why the denominator appears

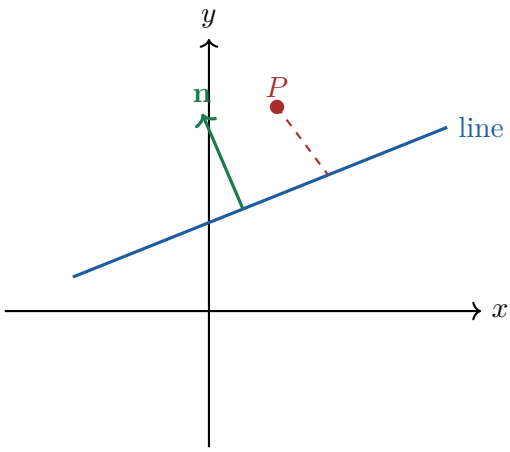
The vector

$$\mathbf{n} = (a, b)$$

is perpendicular to the line $ax + by + c = 0$. It is called a normal vector. The denominator

$$\sqrt{a^2 + b^2} = \|\mathbf{n}\|$$

normalizes the measurement.



Common mistake

Do not forget the absolute value. Distance cannot be negative.

10 Projections

10.1 What is a projection?

Projection answers the question:

> How much of one vector lies in the direction of another vector?

This is important in graphics, machine learning, physics engines, and geometric algorithms.

10.2 Scalar projection

The scalar projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\mathrm{comp}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|}.$$

It is a number.

10.3 Vector projection

The vector projection of $\mathbf{u}$ onto $\mathbf{v}$ is

$$\boxed{\mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}}\mathbf{v}.}$$

It is a vector pointing along $\mathbf{v}$.

Example

Let

$$\mathbf{u} = (3, 4), \qquad \mathbf{v} = (1, 0).$$

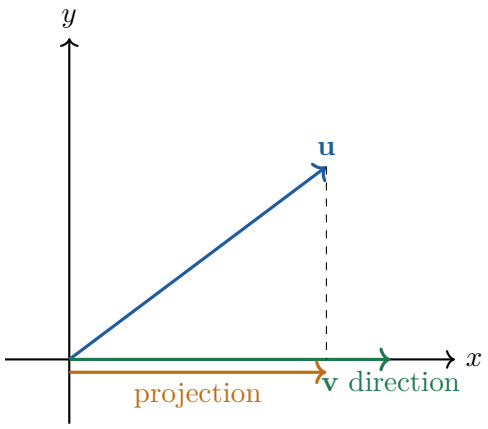
Then

$$\mathbf{u} \cdot \mathbf{v} = 3, \qquad \mathbf{v} \cdot \mathbf{v} = 1.$$

So

$$\mathrm{proj}_{\mathbf{v}}(\mathbf{u}) = 3(1, 0) = (3, 0).$$

This is exactly the horizontal part of $(3, 4)$.



11 Circles

11.1 Definition

A circle is the set of all points at a fixed distance from a fixed point.

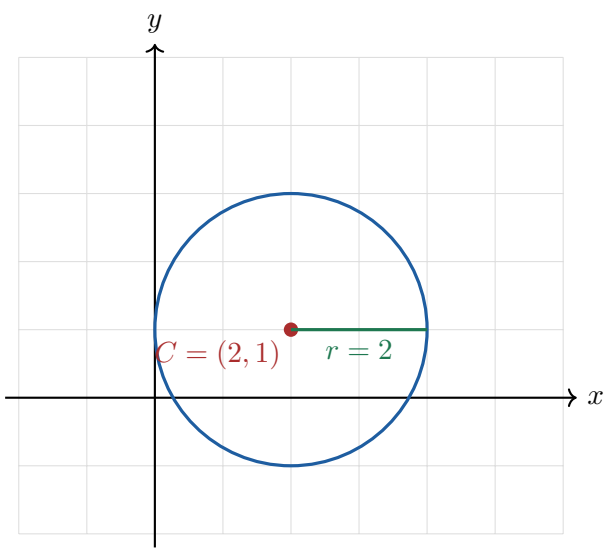
The fixed point is the center. The fixed distance is the radius.

If the center is

$$C = (a, b)$$

and the radius is r, then the circle equation is

$$\boxed{(x-a)^2 + (y-b)^2 = r^2.}$$



Example

The circle with center $(2, 1)$ and radius 3 has equation

$$(x-2)^2 + (y-1)^2 = 9.$$

11.2 Circle centered at the origin

If the center is $(0, 0)$, then the equation becomes

$$\boxed{x^2 + y^2 = r^2.}$$

Example

The equation

$$x^2 + y^2 = 25$$

describes a circle centered at $(0, 0)$ with radius 5.

11.3 Inside, outside, and on the circle

For a circle

$$(x - a)^2 + (y - b)^2 = r^2,$$

compute

$$S = (x - a)^2 + (y - b)^2.$$

Then:

- if $S < r^2$, the point is inside;
- if $S = r^2$, the point is on the circle;
- if $S > r^2$, the point is outside.

Example

Circle:

$$(x - 1)^2 + (y - 2)^2 = 16.$$

Point:

$$P = (4, 6).$$

Compute:

$$S = (4 - 1)^2 + (6 - 2)^2 = 3^2 + 4^2 = 25.$$

Since $25 > 16$, the point is outside the circle.

12 Transformations in the plane

12.1 Translation

A translation moves every point by the same vector.

If a point is

$$P = (x, y)$$

and the translation vector is

$$\mathbf{t} = (a, b),$$

then the new point is

$$P' = (x + a, y + b).$$

Example

Translate $P = (2, 3)$ by $\mathbf{t} = (5, -1)$.

$$P' = (2 + 5, 3 - 1) = (7, 2).$$

12.2 Scaling

Scaling from the origin by factor s sends

$$(x, y) \mapsto (sx, sy).$$

If $s = 2$, objects become twice as large. If $0 < s < 1$, objects shrink. If $s < 0$, the object is also flipped through the origin.

12.3 Rotation

Rotation by angle θ around the origin is given by

$$\boxed{\begin{aligned} x' &= x\cos\theta - y\sin\theta, \\ y' &= x\sin\theta + y\cos\theta. \end{aligned}}$$

For a 90° counterclockwise rotation:

$$\cos 90^\circ = 0, \qquad \sin 90^\circ = 1.$$

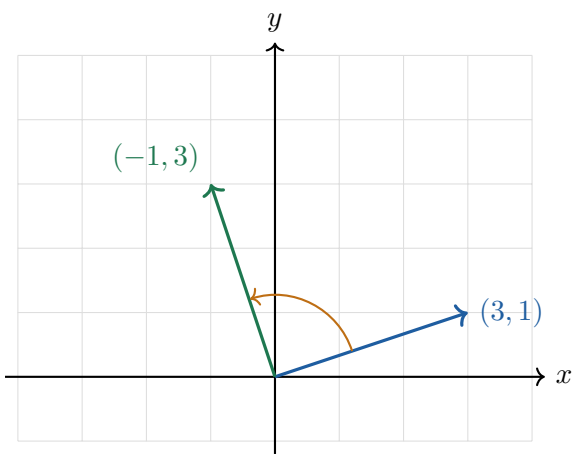
So

$$(x, y) \mapsto (-y, x).$$

Example

Rotate $P = (3, 1)$ by 90° counterclockwise.

$$P' = (-1, 3).$$



12.4 Matrix form

Transformations can be written with matrices. For example, rotation is

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

For 90°:

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Key idea

In computer graphics, matrices are not abstract decoration. They are the standard way to implement transformations.

13 Triangles and orientation

13.1 Area of a triangle from coordinates

For points

$$A = (x_1, y_1), \quad B = (x_2, y_2), \quad C = (x_3, y_3),$$

the area of triangle ABC is

$$\boxed{\text{Area} = \frac{1}{2}\left|x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)\right|.}$$

Example

Find the area of the triangle with

$$A = (0, 0), \quad B = (4, 0), \quad C = (1, 3).$$

Using the formula:

$$\text{Area} = \frac{1}{2}\left|0(0-3) + 4(3-0) + 1(0-0)\right| = \frac{1}{2} \cdot 12 = 6.$$

13.2 Orientation test

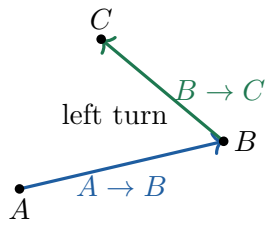
For algorithms, it is useful to know whether three points make a left turn, right turn, or are collinear.

Define

$$D = (x_2 - x_1)(y_3 - y_1) - (y_2 - y_1)(x_3 - x_1).$$

Then:

- if $D > 0$, the turn from A to B to C is counterclockwise;
- if $D < 0$, the turn is clockwise;
- if $D = 0$, the points are collinear.



Key idea

The orientation test is a tiny formula behind many geometry algorithms: convex hulls, polygon tests, segment intersection, and triangulation.

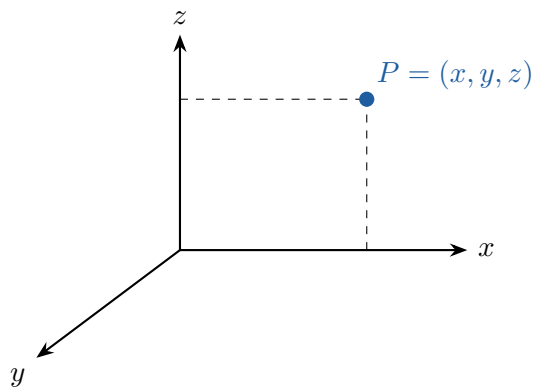
14 Three-dimensional coordinates

14.1 Points in 3D

In three-dimensional space, a point has three coordinates:

$$P = (x, y, z).$$

The coordinates mean movement along three perpendicular axes.



14.2 Distance in 3D

For

$$A = (x_1, y_1, z_1), \qquad B = (x_2, y_2, z_2),$$

the distance is

$$\boxed{d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.}$$

Example

Find the distance between $A = (1, 2, 3)$ and $B = (5, 2, 6)$.

$$d = \sqrt{(5-1)^2 + (2-2)^2 + (6-3)^2} = \sqrt{16 + 0 + 9} = 5.$$

14.3 Vectors in 3D

A 3D vector is

$$\mathbf{v} = (a, b, c).$$

Its length is

$$\|\mathbf{v}\| = \sqrt{a^2 + b^2 + c^2}.$$

The dot product is

$$(a, b, c) \cdot (d, e, f) = ad + be + cf.$$

15 Lines in 3D

15.1 Parametric equation of a line

A line in 3D is described by:

- one point on the line;
- one direction vector.

If the point is

$$P_0 = (x_0, y_0, z_0)$$

and the direction vector is

$$\mathbf{v} = (a, b, c),$$

then the line is

$$\boxed{(x, y, z) = (x_0, y_0, z_0) + t(a, b, c), \qquad t \in \mathbb{R}.}$$

In coordinates:

$$\boxed{\begin{aligned} x &= x_0 + ta, \\ y &= y_0 + tb, \\ z &= z_0 + tc. \end{aligned}}$$

Example

Line through $P_0 = (1, 2, 0)$ with direction $\mathbf{v} = (3, -1, 4)$:

$$(x, y, z) = (1, 2, 0) + t(3, -1, 4).$$

So

$$x = 1 + 3t, \qquad y = 2 - t, \qquad z = 4t.$$

Key idea

The parameter t is like a slider. As t changes, the point moves along the line.

16 Planes in 3D

16.1 Normal vector and plane equation

A plane in 3D can be described using a normal vector, which is perpendicular to the plane.

If the normal vector is

$$\mathbf{n} = (a, b, c),$$

then the plane equation has the form

$$\boxed{ax + by + cz + d = 0.}$$

Example

The plane

$$2x - y + 3z - 6 = 0$$

has normal vector

$$\mathbf{n} = (2, -1, 3).$$

16.2 Plane through a point with a normal vector

If a plane passes through

$$P_0 = (x_0, y_0, z_0)$$

and has normal vector

$$\mathbf{n} = (a, b, c),$$

then

$$\boxed{a(x - x_0) + b(y - y_0) + c(z - z_0) = 0.}$$

Example

Find the plane through $P_0 = (1, 2, 3)$ with normal vector $\mathbf{n} = (2, -1, 4)$.

$$2(x - 1) - 1(y - 2) + 4(z - 3) = 0.$$

Simplify:

$$2x - 2 - y + 2 + 4z - 12 = 0,$$

$$2x - y + 4z - 12 = 0.$$

17 Cross product: a vector perpendicular to two vectors

17.1 Definition

For vectors in 3D,

$$\mathbf{u} = (u_1, u_2, u_3), \qquad \mathbf{v} = (v_1, v_2, v_3),$$

the cross product is

$$\boxed{\mathbf{u} \times \mathbf{v} = (u_2v_3 - u_3v_2,\ u_3v_1 - u_1v_3,\ u_1v_2 - u_2v_1).}$$

The result is a vector perpendicular to both $\mathbf{u}$ and $\mathbf{v}$.

Example

Let

$$\mathbf{u} = (1, 0, 0), \qquad \mathbf{v} = (0, 1, 0).$$

Then

$$\mathbf{u} \times \mathbf{v} = (0, 0, 1).$$

This is the positive z direction.

17.2 Finding a plane from three points

To find a plane through three non-collinear points A, B, C:

1. Form two vectors: $\overrightarrow{AB}$ and $\overrightarrow{AC}$.
2. Compute $\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC}$.
3. Use the point-normal equation of the plane.

Example

Let

$$A = (0, 0, 0), \quad B = (1, 0, 0), \quad C = (0, 2, 0).$$

Then

$$\overrightarrow{AB} = (1, 0, 0), \qquad \overrightarrow{AC} = (0, 2, 0).$$

Cross product:

$$\overrightarrow{AB} \times \overrightarrow{AC} = (0, 0, 2).$$

A normal vector is therefore $(0, 0, 1)$. The plane through the origin with normal $(0, 0, 1)$ is

$$z = 0.$$

18 Analytic geometry and linear systems

18.1 A line intersection is a system

Consider two lines:

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

Solving this system means finding a point (x, y) that lies on both lines.

Add the equations:

$$3x = 6,$$

so

$$x = 2.$$

Then

$$2 - y? \quad \text{No. Use } x - y = 1.$$

Substitute $x = 2$:

$$2 - y = 1,$$

so

$$y = 1.$$

The intersection is $(2, 1)$.

Common mistake

When solving systems, always check the final point in every equation. For $(2, 1)$:

$$2 \cdot 2 + 1 = 5, \qquad 2 - 1 = 1.$$

Both equations work.

18.2 Planes and systems in 3D

A system of three linear equations in three variables often represents the intersection of three planes:

$$\begin{cases} a_1x + b_1y + c_1z = d_1, \\ a_2x + b_2y + c_2z = d_2, \\ a_3x + b_3y + c_3z = d_3. \end{cases}$$

Possible outcomes include:

- one point;
- a line;
- a plane;
- no common point.

This is one reason linear algebra and analytic geometry are deeply connected.

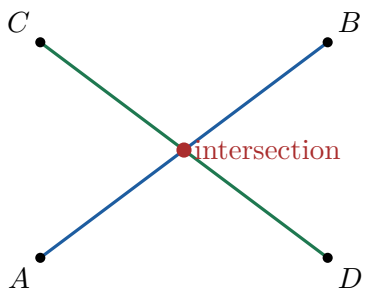
19 Very simple computational geometry

19.1 Segment intersection idea

A line is infinite. A segment is finite. Computer programs usually need segments, not infinite lines.

To test whether two segments intersect, an algorithm often uses orientation tests:

- segment AB intersects segment CD if C and D lie on different sides of line AB, and A and B lie on different sides of line CD;
- special collinear cases must be handled separately.



19.2 Distance-based collision detection

For two circles with centers C_1, C_2 and radii r_1, r_2:

- compute the distance $d(C_1, C_2)$;
- if $d(C_1, C_2) \leq r_1 + r_2$, the circles touch or overlap;
- otherwise, they are separate.

Example

Circle 1 has center $(0, 0)$ and radius 3. Circle 2 has center $(4, 0)$ and radius 2.
The distance between centers is 4. The sum of radii is 5.
Since $4 \leq 5$, the circles overlap.

19.3 Bounding boxes

A simple rectangle aligned with the axes is often called an axis-aligned bounding box.

It is described by four numbers:

$$x_{\min},\ x_{\max},\ y_{\min},\ y_{\max}.$$

A point (x, y) lies inside if

$$x_{\min} \leq x \leq x_{\max}$$

and

$$y_{\min} \leq y \leq y_{\max}.$$

Bounding boxes are used because they are cheap to compute.

20 Numerical caution for programmers

20.1 Floating-point numbers are approximate

In mathematics, we can write $D = 0$ exactly. In a computer program, a computed value might be

$$D = 0.00000000000003$$

instead of 0 because of floating-point rounding.

Therefore, in code we often test

$$|D| < \varepsilon$$

for a small tolerance ε, instead of testing $D = 0$ exactly.

Key idea

Geometry formulas are exact on paper. Implementations are approximate on computers.

20.2 Example: almost collinear points

Suppose three points should lie on one line. The orientation determinant may produce a tiny nonzero number due to rounding. A robust algorithm treats sufficiently small values as zero.

Common mistake

Do not blindly compare floating-point results with zero in geometric algorithms. Use a tolerance when appropriate.

21 Checking whether a point belongs to an object

A very common beginner problem is: given a point and an equation, decide whether the point lies on the object.

The rule is simple:

Key idea

To check whether a point lies on an object, substitute its coordinates into the equation. If the equation becomes true, the point belongs to the object. If not, it does not.

21.1 Point on a line

Example

Does $P = (2, 7)$ lie on the line

$$y = 3x + 1?$$

Substitute $x = 2$ and $y = 7$:

$$7 = 3 \cdot 2 + 1.$$

The right side is

$$6 + 1 = 7.$$

So we get

$$7 = 7.$$

This is true, so P lies on the line.

Example

Does $Q = (1, 5)$ lie on the line

$$2x - y + 4 = 0?$$

Substitute $x = 1$ and $y = 5$:

$$2 \cdot 1 - 5 + 4 = 1.$$

The equation would require this value to be 0. It is not 0, so Q does not lie on the line.

21.2 Point on a circle

Example

Does $P = (4, 5)$ lie on the circle

$$(x-1)^2 + (y-1)^2 = 25?$$

Substitute:

$$(4-1)^2 + (5-1)^2 = 3^2 + 4^2 = 9 + 16 = 25.$$

The right side of the circle equation is also 25. Therefore P lies on the circle.

Common mistake

Substitution is not optional. Guessing from a sketch is dangerous. A drawing is useful for intuition, but the equation gives the exact answer.

22 Parametric lines, rays, and segments in 2D

22.1 Why use parameters?

The equation $y = mx + b$ is useful, but it is not always the most convenient form for programming. In code, we often prefer the following description:

> Start at one point and move in a direction.

This leads to the parametric form of a line.

22.2 Parametric equation of a line

A line through point $P_0 = (x_0, y_0)$ with direction vector $\mathbf{v} = (a, b)$ is

$$\boxed{(x, y) = (x_0, y_0) + t(a, b), \qquad t \in \mathbb{R}.}$$

That means

$$\boxed{x = x_0 + ta, \qquad y = y_0 + tb.}$$

Example

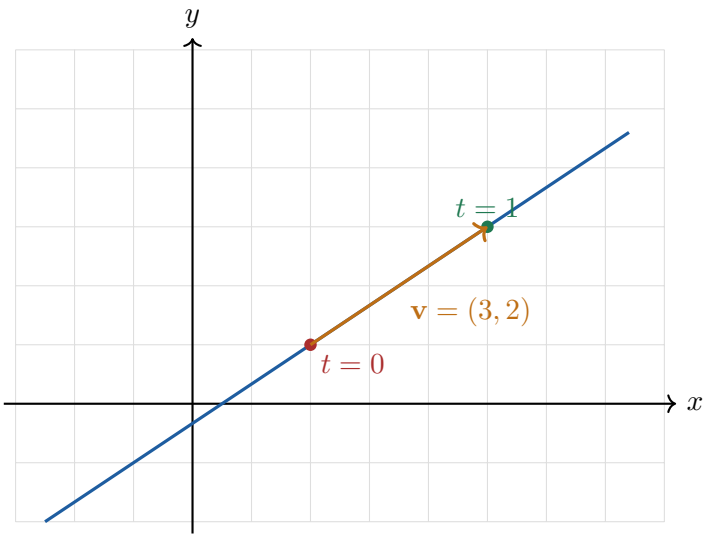
Line through $P_0 = (2, 1)$ with direction $\mathbf{v} = (3, 2)$:

$$(x, y) = (2, 1) + t(3, 2).$$

So

$$x = 2 + 3t, \qquad y = 1 + 2t.$$

If $t = 0$, we get $(2, 1)$. If $t = 1$, we get $(5, 3)$. If $t = -1$, we get $(-1, -1)$.



22.3 Line, ray, segment

The same formula can describe three different objects depending on allowed values of t.

Object	Formula	Allowed parameter values
Line	$P(t) = P_0 + t\mathbf{v}$	$t \in \mathbb{R}$
Ray	$P(t) = P_0 + t\mathbf{v}$	$t \geq 0$
Segment from A to B	$P(t) = A + t(B - A)$	$0 \leq t \leq 1$

Key idea

A line is infinite in both directions. A ray starts somewhere and goes forever in one direction. A segment has two endpoints. In programming, this difference matters a lot.

23 Linear interpolation and points between points

23.1 The formula for a point on a segment

Let A and B be two points. A point between them can be written as

$$\boxed{P(t) = (1 - t)A + tB, \qquad 0 \leq t \leq 1.}$$

This is called linear interpolation, often shortened to *lerp* in computer graphics and game programming.

If $t = 0$, then

$$P(0) = A.$$

If $t = 1$, then

$$P(1) = B.$$

If $t = \frac{1}{2}$, then $P(t)$ is the midpoint.

Example

Let

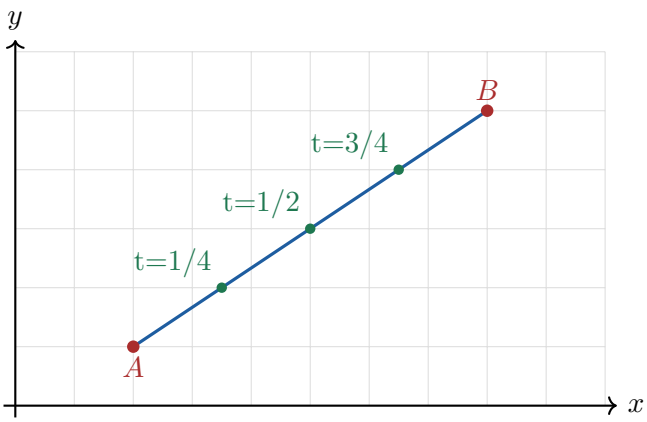
$$A = (2, 1), \qquad B = (8, 5).$$

Find the point one quarter of the way from A to B.
One quarter means $t = \frac{1}{4}$:

$$P\left(\frac{1}{4}\right) = \left(1 - \frac{1}{4}\right)(2, 1) + \frac{1}{4}(8, 5).$$

So

$$P\left(\frac{1}{4}\right) = \frac{3}{4}(2, 1) + \frac{1}{4}(8, 5) = (1.5, 0.75) + (2, 1.25) = (3.5, 2).$$



23.2 Why interpolation matters in computer science

Linear interpolation is used for:

- moving an object smoothly from one position to another;
- blending colors;
- animation;
- sampling points on a segment;
- numerical approximation.

Common mistake

The parameter t is not the same as the x-coordinate. It is a separate number telling us how far along the segment we are.

24 More worked examples: lines slowly

This section repeats line calculations deliberately. Repetition is not a flaw here. Students usually learn lines only after seeing the same operation several times.

Example

Find the line through $(2, 3)$ and $(5, 12)$.
Step 1: slope.

$$m = \frac{12-3}{5-2} = \frac{9}{3} = 3.$$

Step 2: point-slope form using $(2, 3)$.

$$y - 3 = 3(x - 2).$$

Step 3: simplify.

$$y - 3 = 3x - 6,$$

$$y = 3x - 3.$$

Step 4: check with the second point.

$$12 = 3 \cdot 5 - 3 = 15 - 3 = 12.$$

Correct.

Example

Find the line through $(-1, 4)$ and $(3, -4)$.
Slope:

$$m = \frac{-4-4}{3-(-1)} = \frac{-8}{4} = -2.$$

Point-slope form:

$$y - 4 = -2(x + 1).$$

Simplify:

$$y - 4 = -2x - 2,$$

$$y = -2x + 2.$$

Check with $(3, -4)$:

$$-4 = -2 \cdot 3 + 2 = -6 + 2 = -4.$$

Correct.

Example

Find the line through $(4, 2)$ and $(4, 9)$.
The x-coordinates are the same. This is a vertical line.

$$\boxed{x = 4.}$$

Do not try to write $y = mx + b$. A vertical line has no slope.

25 Conics: parabolas and ellipses very briefly

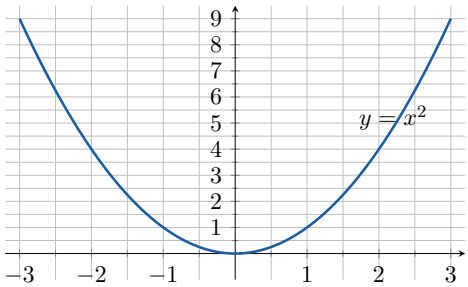
Circles are not the only curves described by equations. A more general family is called conics. For a beginner course, the most important conics to recognize are parabolas and ellipses.

25.1 Parabola

The basic parabola is

$$y = x^2.$$

It is not a line. It curves upward.



Key idea

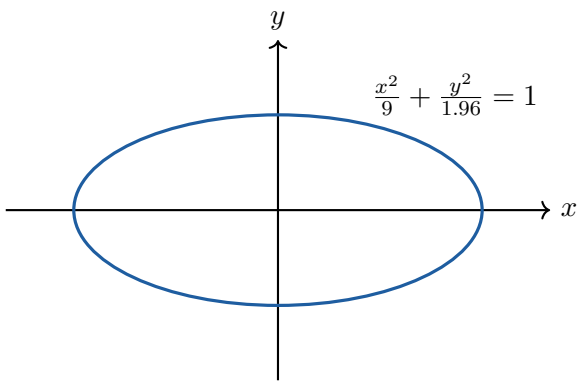
A line has constant slope. A parabola does not. Its steepness changes as x changes.

25.2 Ellipse

An ellipse is a stretched circle. A common equation is

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

If $a = b$, the ellipse becomes a circle.



Common mistake

You do not need to master all conics at once. At the beginning, recognize their equations and understand that equations can describe curves, not only lines.

26 Mini-project: from a word problem to formulas

Suppose a simple game has a circular danger zone.

The danger zone has center

$$C = (10, 5)$$

and radius

$$r = 3.$$

A player is at

$$P = (12, 7).$$

Question: is the player inside the danger zone?

Recipe

Step 1: translate the words into geometry.
Danger zone = circle.
Player position = point.
Step 2: write the circle equation.

$$(x - 10)^2 + (y - 5)^2 = 9.$$

Step 3: substitute player coordinates.

$$(12 - 10)^2 + (7 - 5)^2 = 2^2 + 2^2 = 8.$$

Step 4: compare with $r^2 = 9$.
Since $8 < 9$, the player is inside the circle.

Now suppose the player moves by vector

$$\mathbf{v} = (3, -1).$$

The new position is

$$P' = (12, 7) + (3, -1) = (15, 6).$$

Check again:

$$(15 - 10)^2 + (6 - 5)^2 = 5^2 + 1^2 = 26.$$

Since $26 > 9$, the player is now outside the danger zone.

Key idea

This tiny example already uses points, vectors, circles, substitution, and inequalities. That is the practical power of analytic geometry.

27 Formula sheet

Distance

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Midpoint

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

Vector

$$\overrightarrow{AB} = (x_2 - x_1, y_2 - y_1)$$

$$\|(a,b)\| = \sqrt{a^2+b^2}$$

Dot product

$$(a,b)\cdot(c,d) = ac+bd$$
$$\mathbf{u}\cdot\mathbf{v} = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$$

Line

$$m = \frac{y_2-y_1}{x_2-x_1}$$
$$y-y_0 = m(x-x_0)$$
$$ax+by+c=0$$

Point-line distance

$$d = \frac{|ax_0+by_0+c|}{\sqrt{a^2+b^2}}$$

Circle

$$(x-a)^2+(y-b)^2=r^2$$

Rotation

$$x' = x\cos\theta - y\sin\theta$$
$$y' = x\sin\theta + y\cos\theta$$

28 Practice problems

A. Points, distances, and midpoints

1. Plot the points $(2,3)$, $(-1,4)$, $(-3,-2)$, $(5,-1)$.
2. Find the distance between $(1,2)$ and $(7,10)$.
3. Find the midpoint of $(-4,5)$ and $(8,-1)$.
4. A point moves from $(3,2)$ by vector $(-5,4)$. What is its new position?

B. Vectors

1. Compute $(3,4)+(-1,7)$.
2. Compute $5(2,-3)$.
3. Find the length of $(-6,8)$.
4. Are $(1,2)$ and $(4,-2)$ perpendicular?

C. Lines

1. Find the slope of the line through $(2,1)$ and $(6,9)$.
2. Find the equation of the line with slope -3 through $(1,5)$.
3. Find the equation of the line through $(0,2)$ and $(4,10)$.
4. Are $y=2x+1$ and $y=2x-7$ parallel?
5. Are $y=\frac{1}{2}x+4$ and $y=-2x+1$ perpendicular?

D. Circles and transformations

1. Write the equation of the circle with center $(3,-2)$ and radius 5.
2. Is the point $(6,2)$ inside, outside, or on the circle $(x-1)^2+(y-2)^2=16$?
3. Translate $(4,-1)$ by vector $(-2,6)$.
4. Rotate $(5,2)$ by 90° counterclockwise around the origin.

E. 3D geometry

1. Find the distance between $(1, 0, 2)$ and $(5, 3, 14)$.
2. Write a parametric equation of the line through $(1, 2, 3)$ with direction $(4, 0, -1)$.
3. Find a plane through $(1, 1, 1)$ with normal vector $(2, 3, -1)$.

29 Answers to selected practice problems

A

1. Plotting exercise.
2. $\sqrt{(7-1)^2 + (10-2)^2} = \sqrt{36+64} = 10$.
3. $M = \left(\frac{-4+8}{2}, \frac{5+(-1)}{2}\right) = (2, 2)$.
4. $(3, 2) + (-5, 4) = (-2, 6)$.

B

1. $(3, 4) + (-1, 7) = (2, 11)$.
2. $5(2, -3) = (10, -15)$.
3. $\sqrt{(-6)^2 + 8^2} = 10$.
4. $(1, 2) \cdot (4, -2) = 4 - 4 = 0$, so yes.

C

1. $m = \frac{9-1}{6-2} = 2$.
2. $y - 5 = -3(x - 1)$, so $y = -3x + 8$.
3. Slope $m = 2$, so $y = 2x + 2$.
4. Yes, same slope.
5. Yes, slopes multiply to -1.

D

1. $(x - 3)^2 + (y + 2)^2 = 25$.
2. Compute $(6 - 1)^2 + (2 - 2)^2 = 25 > 16$, so outside.
3. $(4, -1) + (-2, 6) = (2, 5)$.
4. $(5, 2) \mapsto (-2, 5)$.

E

1. $\sqrt{4^2 + 3^2 + 12^2} = 13$.
2. $(x, y, z) = (1, 2, 3) + t(4, 0, -1)$.
3. $2(x - 1) + 3(y - 1) - 1(z - 1) = 0$, or $2x + 3y - z - 4 = 0$.

Final note

Analytic geometry is not a set of isolated formulas. It is a language. Points, vectors, lines, circles, planes, and transformations are the vocabulary. Equations and coordinates are the grammar. Once this language becomes familiar, many topics in computer science become easier: graphics, games, robotics, machine learning, computer vision, numerical methods, and algorithms.

The safest way to learn it is not to memorize everything at once. Instead, practice the same small operations many times:

- plot points;
- compute differences of coordinates;
- build vectors;
- compute distances;
- write equations of lines;
- solve simple intersections;
- check answers by substitution.

Small correct steps are enough. Analytic geometry becomes difficult mostly when students skip the basics.